$$Number\ Of\ Frames = \frac{(Input\ Length - 1) + (2 * Frame\ Size)}{Frame\ Size}$$

$$Let:$$

$$Number\ Of\ Frames\ =\ N$$

$$Input\ Length\ =\ i$$

$$Frame\ Size\ =\ \mathrm{x}$$

$$N = \frac{(i-1) + (2 * x)}{x}$$

$$N = \frac{i-1}{x} + \frac{2x}{x}$$

$$N = \frac{i-1}{x} + 2$$

$$N - 2 = \frac{i-1}{x}$$

$$x(N-2) = i - 1$$

$$x(N-2) + 1 = i$$

$$\therefore Input\ Length = Frame\ Size(Number\ Of\ Frames - 2) + 1$$